BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Practical Programming Project

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. For valid ages 13-18, give an ordinary normal test value.

2. Where can a local variable be used?
   A) In every program
   B) Within its local scope
   C) Across every network
   D) In all other subprograms

3. "Data interception is only copying a paper document from a filing cabinet." Correct this. Explain why.

4. A new user needs to open and sort files. "A user interface replaces all CPU instructions." Correct this. How does an interface help?

Today's objectives

Design, implement and test a small working program. Apply this to another software scenario.

Key vocabulary - Read, cover, recall

Design
Define success criteria, inputs, algorithms and expected outputs before writing code.

Implementation
Build a small working feature using appropriate constructs.

Testing
Record test inputs, expected outputs, actual outcomes and pass/fail results.

Refinement
Use test evidence to correct program faults and improve reliability.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Design
Define success criteria, inputs, algorithms and expected outputs before writing code. Consider this in a ticket-sales mini project.

Implementation
Build a small working feature using appropriate constructs. Consider this in a pupil quiz project.

Testing
Record test inputs, expected outputs, actual outcomes and pass/fail results. Consider this in a school lunch calculator.

Refinement
Use test evidence to correct program faults and improve reliability. Consider this in a sports points tracker.

I Do - Worked model

A ticket-sales mini project needs design and implementation. Explain why. [4 marks]

- Identify the inputs, constraints and expected result.
- Design [1]; Define success criteria, inputs, algorithms and expected outputs before writing code. [1].
- Implementation [1]; Build a small working feature using appropriate constructs. [1].
- Give a specific point and a linked reason for each feature.

We Do - Guided practice

1. In a pupil quiz project, explain the purpose of design. [2 marks]

2. A school lunch calculator needs implementation. Explain its role. [2 marks]

3. A sports points tracker uses testing and refinement. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Trace the OCR ERL program, test a new input and improve the code.

```
ticketPrice = 5
quantity = 3
total = ticketPrice * quantity
print(total)
```

You Do - Independent exam questions

Q2. State TWO concepts relevant to this topic. [2 marks]

Q3. Explain how implementation and testing help in a book-return reminder. [4 marks]

Q4. Explain the roles of design, testing and refinement in a classroom club booking. [6 marks]

Exit Ticket - True or False?

Planning a program involves identifying success criteria, inputs, algorithms and expected outputs before coding.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. 15 is a representative valid value away from the boundaries.

2. B - Within its local scope

3. Capturing data during transmission, potentially exposing confidential information.

4. A user interface lets people use the computer, through GUI controls or typed commands. Other OS tasks manage devices, accounts, memory and files.

I Do - Explain the model

Identify the inputs, constraints and expected result.

Design [1]; Define success criteria, inputs, algorithms and expected outputs before writing code. [1].

Implementation [1]; Build a small working feature using appropriate constructs. [1].

Give a specific point and a linked reason for each feature.

We Do - Model answers

1. Design [1]; Define success criteria, inputs, algorithms and expected outputs before writing code. [1].

2. Implementation [1]; Build a small working feature using appropriate constructs. [1].

3. Testing [1]; Record test inputs, expected outputs, actual outcomes and pass/fail results. [1]. Refinement [1]; Use test evidence to correct program faults and improve reliability. [1].

Code First - Answer

Original output: 15.
Updated tests: At 0: total 0; at 4: total 20. A negative value should print an error after validation is added.
Write the expected and actual output separately.

You Do - Self-mark

Q2: Design [1]; Testing [1].

Q3: Implementation [1]; Build a small working feature using appropriate constructs. [1]. Testing [1]; Record test inputs, expected outputs, actual outcomes and pass/fail results. [1].

Q4: Design [1]; Define success criteria, inputs, algorithms and expected outputs before writing code. [1]. Testing [1]; Record test inputs, expected outputs, actual outcomes and pass/fail results. [1]. Refinement [1]; Use test evidence to correct program faults and improve reliability. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.